

Singularity-Free Discrete Global Grid Boundaries: Cartesian 3D Vertex Extraction for Interfacial Mass, Enthalpy, and Exergy Dissipation Monads

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Abstract

Planetary-scale ecological and biogeochemical simulations require discrete spatial formulations that strictly preserve the First and Second Laws of Thermodynamics. When global dynamics are discretized over Discrete Global Grid Systems (DGGS) using angular geodetic coordinates (ϕ, λ) , numerical artifacts arise: metric singularities at the poles, latitude-dependent distortion, and asymmetric interfacial flux evaluations that violate conservation laws. In Sprint 069 of the Web of Life simulation architecture, we developed and verified `extractH3BoundaryCartesianVertices3D` within the spatial adjacency subsystem. This module transforms geodetic hexagonal and pentagonal boundary loops into normalized 3D Cartesian coordinates $\mathbf{v} \in \mathbb{S}^2 \subset \mathbb{R}^3$. We derive the differential geometric framework for interfacial tangent vectors, surface normal evaluation, and geodesic edge lengths. We demonstrate how this Cartesian projection ensures exact boundary anti-symmetry ($\hat{\mathbf{n}}_{AB} = -\hat{\mathbf{n}}_{BA}$), eliminates polar coordinate singularities, and guarantees machine-level conservation of interfacial mass, enthalpy, and positive-definite entropy generation across the planetary discrete biosphere.

1 Introduction and Physical Motivation

The Earth’s biosphere is a dissipative, non-equilibrium thermodynamic system sustained by a continuous flux of low-entropy solar exergy and the radiative rejection of high-entropy heat to space [1]. Modeling the macro-scale self-organization of life, atmospheric vapor advection, ocean carbon circulation, and continental nutrient transport requires partition schemes that balance spatial uniformity with rigorous geometric conservation [2].

The Web of Life simulation framework (<https://github.com/pascalranoroarijaona/WebOfLife>) discretizes planetary surfaces using the Uber H3 discrete global grid system, an icosahedral aperture-7 hexagonal and pentagonal tessellation. In finite-volume modeling over such grids, state transitions in each cell i are determined by the summation of fluxes across its cell interfaces Γ_{ij} with neighboring cells $j \in \mathcal{N}(i)$:

$$\frac{d\mathbf{S}_i}{dt} = \sum_{j \in \mathcal{N}(i)} \Phi_{ij} + \Pi_i \quad (1)$$

where $\mathbf{S}_i$ denotes the conserved stock tensor (composed of water mass $M_{\text{H}_2\text{O}}$, carbon M_{C} , oxygen M_{O_2} , trace minerals M_{min} , and internal thermal energy U), Φ_{ij} is the net flux across interface Γ_{ij} , and Π_i accounts for internal biogeochemical transformations.

Prior to Sprint 069, DGGS interfaces were parameterized by spherical geodetic latitude and longitude coordinates (ϕ, λ) . In this conventional parameterization, the Riemannian metric

tensor:

$$g_{\mu\nu} = \begin{bmatrix} R^2 & 0 \\ 0 & R^2 \cos^2 \phi \end{bmatrix} \quad (2)$$

degenerates at the planetary poles ($\phi \rightarrow \pm\pi/2$), resulting in vanishing metric determinants, distorted distance approximations, and asymmetric flux estimates between polar cells. By mapping boundary vertices directly into Cartesian unit 3-vectors on the unit 2-sphere $\mathbb{S}^2 \subset \mathbb{R}^3$, coordinate singularities are eliminated, and interface geometries become strictly invariant under planetary rotation.

2 Mathematical Formulation of Cartesian Boundary Projection

2.1 Geodetic to Spherical Cartesian Mapping

Let an H3 cell boundary be defined by a sequence of N geodetic vertices in radians, (ϕ_k, λ_k) for $k \in \{0, \dots, N-1\}$, where $N = 6$ for hexagonal cells and $N = 5$ for the twelve topological pentagons. For a sphere of radius R , the Cartesian embedding vector $\mathbf{v}_k = [x_k, y_k, z_k]^T$ is defined by:

$$\begin{aligned} x_k &= R \cos(\phi_k) \cos(\lambda_k) \\ y_k &= R \cos(\phi_k) \sin(\lambda_k) \\ z_k &= R \sin(\phi_k) \end{aligned} \quad (3)$$

For default unit-sphere calculations ($R = 1.0$), every vertex obeys the Euclidean norm invariant:

$$\|\mathbf{v}_k\|_2 = \sqrt{x_k^2 + y_k^2 + z_k^2} = 1.0 \pm \varepsilon_{\text{machine}} \quad (4)$$

where $\varepsilon_{\text{machine}} < 10^{-15}$ under double-precision IEEE-754 arithmetic.

2.2 Centroid Normalization and Loop Topology

Boundary sequences are oriented in a counter-clockwise (CCW) winding order when viewed from an exterior observation point along the outward radial vector. To provide closed-loop compatibility for graphic rendering and Stokes theorem path integrations, the sequence can optionally be extended to $N + 1$ points such that $\mathbf{v}_N = \mathbf{v}_0$.

The geometric boundary centroid $\mathbf{c} \in \mathbb{R}^3$ is derived from the unweighted arithmetic mean of the boundary vertices, projected radially onto the spherical manifold:

$$\bar{\mathbf{c}} = \frac{1}{N} \sum_{k=0}^{N-1} \mathbf{v}_k, \quad \mathbf{c} = R \frac{\bar{\mathbf{c}}}{\|\bar{\mathbf{c}}\|_2} \quad (5)$$

This formulation avoids polar trigonometric singularities and exhibits collinear alignment with the native geodetic cell center within an angular discrepancy tolerance of $\theta < 0.05$ rad.

3 Interfacial Fluid, Heat, and Exergy Dynamics

3.1 Interface Metrics from Cartesian Boundary Pairs

Consider two topologically adjacent H3 cells A and B . Their shared spatial boundary Γ_{AB} is an arc of a great circle delimited by two Cartesian vertices $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{S}^2$:

1. Directed Interface Segment:

$$\mathbf{e}_{AB} = \mathbf{v}_2 - \mathbf{v}_1 \quad (6)$$

2. Geodesic Arc Length:

$$L_{AB} = R \arccos(\text{clamp}(\mathbf{v}_1 \cdot \mathbf{v}_2, -1, 1)) \quad (7)$$

3. Normalized Midpoint Vector:

$$\mathbf{m}_{AB} = \frac{\mathbf{v}_1 + \mathbf{v}_2}{\|\mathbf{v}_1 + \mathbf{v}_2\|_2} \quad (8)$$

4. Outward Interface Unit Normal:

$$\hat{\mathbf{n}}_{AB} = \frac{\mathbf{e}_{AB} \times \mathbf{m}_{AB}}{\|\mathbf{e}_{AB} \times \mathbf{m}_{AB}\|_2} \quad (9)$$

Because $\mathbf{e}_{BA} = \mathbf{v}_1 - \mathbf{v}_2 = -\mathbf{e}_{AB}$ and $\mathbf{m}_{BA} = \mathbf{m}_{AB}$, it follows analytically that:

$$\hat{\mathbf{n}}_{BA} = \frac{(-\mathbf{e}_{AB}) \times \mathbf{m}_{AB}}{\|(-\mathbf{e}_{AB}) \times \mathbf{m}_{AB}\|_2} = -\hat{\mathbf{n}}_{AB} \quad (10)$$

Equation (10) guarantees machine-level anti-symmetry of edge normals without ad-hoc sign corrections.

3.2 First Law Conservative Stock Transfers

Let the fluid velocity vector evaluated at midpoint $\mathbf{m}_{AB}$ be $\mathbf{u}_{AB} \in \mathbb{R}^3$. The directed normal flux velocity is $u_{n,AB} = \mathbf{u}_{AB} \cdot \hat{\mathbf{n}}_{AB}$. For an effective vertical layer depth Δz_{eff} , the interfacial cross-sectional area is $A_{AB} = L_{AB} \Delta z_{\text{eff}}$, yielding the volumetric flux rate:

$$Q_{AB} = u_{n,AB} A_{AB} \quad (11)$$

Advective mass transfer for conserved chemical species k over discrete time step Δt is discretized using an upwind donor-cell scheme:

$$\Delta M_{k,AB} = Q_{AB} [C]_{k,\text{Donor}} \Delta t, \quad \text{Donor} = \begin{cases} A, & Q_{AB} \geq 0 \\ B, & Q_{AB} < 0 \end{cases} \quad (12)$$

where $[C]_k$ is the volumetric concentration of species k . Because $u_{n,BA} = -u_{n,AB}$, the exchange satisfies strict First Law mass conservation:

$$\Delta M_{k,AB} + \Delta M_{k,BA} = 0 \quad (13)$$

3.3 Second Law Irreversible Entropy Generation

Thermal enthalpy transfer combines advective transport and Fourier conduction:

$$\Phi_{U,AB} = Q_{AB} c_p \rho T_{\text{Donor}} + \kappa A_{AB} \frac{T_A - T_B}{d(\mathbf{c}_A, \mathbf{c}_B)} \quad (14)$$

where c_p is isobaric heat capacity, κ is thermal conductivity, and $d(\mathbf{c}_A, \mathbf{c}_B) = R \arccos(\mathbf{c}_A \cdot \mathbf{c}_B)$ is the geodesic distance between cell centroids. The entropy generation rate associated with interfacial heat conduction is:

$$\dot{S}_{\text{conduction},AB} = \kappa A_{AB} \frac{T_A - T_B}{d(\mathbf{c}_A, \mathbf{c}_B)} \left(\frac{1}{T_B} - \frac{1}{T_A} \right) = \kappa A_{AB} \frac{(T_A - T_B)^2}{d(\mathbf{c}_A, \mathbf{c}_B) T_A T_B} \geq 0 \quad (15)$$

By computing A_{AB} and $d(\mathbf{c}_A, \mathbf{c}_B)$ via Cartesian dot products, the simulation guarantees that $\dot{S}_{\text{conduction},AB} \geq 0$ under all thermal configurations, satisfying the Second Law of Thermodynamics without polar metric contamination.

4 Implementation Architecture in TypeScript

The transformation pipeline is implemented in `src/spatial/h3_adjacency.ts` within the Web of Life core engine. The interface specification is defined as follows:

```
export interface Cartesian3D {
  readonly x: number;
  readonly y: number;
  readonly z: number;
}

export interface H3BoundaryCartesian3D {
  readonly h3Index: string;
  readonly vertexCount: number;
  readonly vertices: readonly Cartesian3D[];
  readonly isClosed: boolean;
  readonly centroid: Cartesian3D;
}
```

The primary function `extractH3BoundaryCartesianVertices3D` performs the trigonometric conversion in a single forward pass, calculating the 3D unit coordinates, enforcing immutability through frozen object structures, and normalizing the resulting centroid vector.

5 Empirical Verification and Invariant Analysis

The implementation was validated across test suites covering H3 resolutions $r \in [0, 8]$ on both hexagonal cells and pentagonal singular cells. Five physical and geometric invariants were verified:

Invariant ID	Physical/Geometric Principle	Mathematical Formulation	Status
INV-G-01	Radial Manifold Embedding	$ \mathbf{v}_k _2 - 1.0 < 10^{-12}$	Verified
INV-G-02	Topological Vertex Parity	$N \in \{5, 6\}$, closed length = $N + 1$	Verified
INV-G-03	Centroid Angular Consistency	$\arccos(\mathbf{c} \cdot \mathbf{c}_{\text{center}}) < 0.05 \text{ rad}$	Verified
INV-M-04	Interfacial Normal Anti-Symmetry	$ \hat{\mathbf{n}}_{AB} + \hat{\mathbf{n}}_{BA} _2 < 10^{-14}$	Verified
INV-T-05	Non-Negative Entropy Dissipation	$\Delta S_{\text{entropy}, AB} \geq 0$	Verified

Table 1: Invariant validation matrix executed during Sprint 069 test cycles.

6 Significance for Systems Ecology and Planetary Modeling

In computational ecology, inaccurate interfacial geometry induces artificial dispersion and numerical dissipation, which obscures ecological phenomena such as boundary-layer trophic gradients, marine oxygen minimum zones, and desertification frontiers [3]. By grounding the discrete global grid system in Cartesian 3D boundary geometry, the Web of Life simulation architecture ensures:

1. **Exact Mass Conservation:** Advective transport of abiotic stocks across cell boundaries creates zero artificial source/sink errors, allowing closed-loop ecological biogeochemical cycles.
2. **Rigorous Exergy Tracking:** Metabolic dissipation and abiotic entropy production can be cleanly separated from numerical discretization errors.

3. **Rendering and Simulation Cohesion:** Cartesian boundary vectors can be ingested directly by WebGL and WebGPU shader pipelines without re-projection overhead.

7 Conclusion and Future Work

Sprint 069 establishes a coordinate-singularity-free foundation for interfacial transport across the Web of Life simulation framework. Transforming H3 boundary loops into 3D Cartesian coordinates preserves metric anti-symmetry and enforces fundamental thermodynamic laws at the discrete level. Upcoming sprints will integrate these Cartesian boundary metrics into the full `SpatialFluxMonad`, enabling multi-layer baroclinic fluid transport and reactive biological dispersal across the global planetary grid.

Code Availability

The full implementation, test specifications, and architectural documentation are available in the official GitHub repository:

<https://github.com/pascalranoroarijaona/WebOfLife>

References

- [1] A. Kleidon, *Life, hierarchy, and the thermodynamics of Earth system process*, Physics of Life Reviews, 7(4):424–460, 2010.
- [2] K. Sahr, D. White, and A. J. Kimerling, *Geodesic discrete global grid systems*, Cartography and Geographic Information Science, 30(2):121–134, 2003.
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